

Thaneesan Siventhirarajah

📍 Jaffna, Sri Lanka.

☎ +94 705951199

✉ thaneesan11@gmail.com

🌐 LinkedIn

🔍 ResearchGate

🌐 Personal Website

SUMMARY

Mechatronics graduate with research and hands-on development experience in robotics, computer vision, additive manufacturing, digital fabrication, and intelligent systems. Experienced in developing and prototyping research-driven solutions using CAD/CAM, machine learning, embedded systems, and rapid prototyping. Interested in pursuing research in robotics, intelligent systems, assistive and wearable technologies, and advanced digital manufacturing.

EDUCATION

BTech Hons (Mechatronics) (2020 - 2024)

*Bachelor of Technology Honours in Science and Technology specialized in Mechatronics,
Faculty of Applied Sciences of Uva Wellassa University of Sri Lanka.*

- GPA - 3.72 / 4.00 (First Class Honours)
- Medium of Instruction - English
- Effective Date of the Degree - 30 September 2024
- Programme Duration - 4 Years

Developing Career Skills (2018)

Department of Technical Education & Training Sri Lanka, Ministry of Skills Development & Vocational Training

G.C.E (A/L) Physical Science Stream (2018)

Jaffna Hindu College

Z-Score - 1.2306

Combined Mathematics - B | Physics - B | Chemistry - B

G.C.E (O/L) (2015)

Jaffna Hindu College

Mathematics - A | Science - B | English - B | ICT - A

RESEARCH INTERESTS

- Robotics & Intelligent Systems
- Human-Centered / Assistive Robotics
- Control Systems & Mechatronics modelling
- Product Design and Development
- Advanced Kinematics
- Soft Robotics & Wearable Technologies
- Additive Manufacturing & Digital Fabrication
- Computer Vision & Machine Learning

PUBLICATIONS

- Thaneesan S., & Jayasinghe, J. A. K. S. (2024). YOLOv8 Powered Solutions for Box Identification in Warehouses. *International Journal of Innovative Science and Research Technology (IJISRT)*, 1560–1565. <https://doi.org/10.38124/IJISRT/IJISRT24SEP1017>
- Thaneesan Siventhirarajah, D.D.B. Senanayake, M.P.A.M. Rathnakumara, & R.D.A.U. Pallegama. (2024). *Voice-controlled prosthetic hand with keyword spotting*. Conference: IRCUWU2024
- Thaneesan Siventhirarajah, & N. Gajaruban. (2025). *Integrating Contemporary Casting Technologies in Traditional Goldsmithing A Case Study on Gold Decker Ring Production*. 3rd GJRTI International Research Symposium

PROJECTS

Warehouse Box Identification Using Computer Vision

- Investigated computer vision and deep-learning approaches, including YOLO and U-Net, for automated box identification.
- Explored machine-learning-based solutions for warehouse automation.
- Technologies: Python, YOLOv8, Computer Vision, Machine Learning

Voice-Controlled Prosthetic Hand with Keyword Spotting

- Developed a voice-controlled prosthetic-hand concept integrating keyword spotting, machine learning, CAD modelling, and prototype development.
- Technologies: Python, ATmega328P microcontroller, CNN, NLP, CAD, Prototyping

Obstacle Detection System for Visual Assistance

- Developed a real-time obstacle-detection system using TFLite and Raspberry Pi.
- Applied computer vision and machine-learning techniques to support visual assistance applications.
- Technologies: YOLOv8, TensorFlow, Raspberry Pi, Computer Vision

Smart Chair Walking Stick

- Developed a prototype combining CAD modelling, embedded systems, and IoT for an assistive application.
- Technologies: ESP32, IoT, CAD, Prototyping

Inclined Bucket Conveyor Belt System

- Designed and developed a prototype inclined bucket conveyor system using CAD modelling and Arduino-based control.
- Technologies: CAD Modelling, Arduino, Prototyping

Knight Rider circuit using a PIC microcontroller

- Circuit design, microcontroller programming and embedded system fundamentals.
- Technologies: PIC microcontroller, Proteus

EXPERIENCE

Product Development & Additive Manufacturing

Tensor Labs

2024 - 2026

Applied 3D printing and digital fabrication techniques for rapid prototyping and functional product development.

R&D – Digital Jewellery Manufacturing

Ruby Jewellery (Pvt) Ltd.

May 2025 - Present

SLA resin 3D printing, CAD-based design, investment casting, fiber laser cutting and engraving, and laser spot welding.

Independent Researcher

Self-employed – Part-time

2025 - Present

Delivered projects in robotics, computer vision, CAD design, and additive manufacturing.

Research Assistant Intern

LE Robotics (Pvt) Ltd.

June 2024 - September 2024

Worked on a computer vision project focused on warehouse applications.

STEM Workshop Instructor

2022 - Present

Conduct STEM workshops for schools and community events, introducing students to robotics, electronics, programming, and engineering fundamentals.

CERTIFICATIONS

- SOLIDWORKS CAD Design Associate (CSWA) - Dassault Systèmes
- Additive Manufacturing: Resin 3D Printing: LinkedIn Learning
- Rapid Prototyping for Product Design: LinkedIn Learning
- Modeling and Simulation with Simulink: Coursera
- Simulink Onramp: MathWorks

- MATLAB Onramp: MathWorks
- SOLIDWORKS Sheet Metal: Udemy
- SOLIDWORKS: Become a Certified Professional Today (CSWP): Udemy
- Introduction to CNC Programming: Udemy

TECHNICAL SKILLS

- **Programming:** Python, C, C++, MATLAB
- **AI & Computer Vision:** YOLO, TensorFlow, Computer Vision, Machine Learning
- **CAD/CAM:** SOLIDWORKS, Rhino, AutoCAD, Meshmixer, CAD Modeling, CAM
- **Prototyping & Fabrication:** FDM/SLA 3D Printing, Rapid Prototyping, Laser Cutting/Engraving, Investment Casting
- **Embedded Systems:** Arduino, ESP32, Raspberry Pi, IoT
- **Engineering:** Mechatronics, Control Systems, Digital Fabrication, Product Development

LANGUAGE

English (Professional Working)

Tamil (Native)

Sinhala (Elementary)

HONORS & AWARDS

- **Dean's List for Outstanding Academic Performance — 5 semesters**
- Skilled youth Entrepreneurship competition - 2024
- LinkedEng English Camp 2021 (Participation in the Speech contest and Debate competition)
- X-mas Revamp (The most creative Category 2nd Place)

REFERENCES

Prof. J. A. K. S. Jayasinghe

Professor Emeritus,

*Department of Electronic & Telecommunication Engineering, Faculty of Engineering,
University of Moratuwa, Sri Lanka.*

Email: jaks@uom.lk

Phone: +94777688894

Relationship: Research Assistant Supervisor

Dr. A.R.P.C.C.J. Amarasinghe

Senior Lecturer,

*Department of Science and Technology, Faculty of Applied Sciences of
Uva Wellassa University of Sri Lanka.*

Email: chinthakaa@uwu.ac.lk

Phone: +94715104687

Relationship: Undergraduate Lecturer

Duranga Senanayake

PhD Candidate

*Institute of Innovation, Science and Sustainability
Federation University Australia.*

Email: d.senanayake@federation.edu.au

Phone: +94766844268

Relationship: Undergraduate Research Supervisor

Thevathayarajh Thayananthan

Ph.D. Candidate and Graduate Research Assistant

*School of Environmental, Civil, Agricultural, and Mechanical Engineering
College of Engineering, University of Georgia.*

Email: theva@uga.edu

Phone: +1 6623706844

Relationship: Undergraduate Lecturer